

CO3 AT1 OUTPUTS

1.

```
er: '49804' -- 'C:\Users\reddy\nlp\DSA0304-NATURAL-LANGUAGE-PROCESSING\CO3 AT1\Q1.PY'
Enter N (1, 2 or 3): 1

Unigram Counts:
Counter({'the': 5, 'is': 5, 'student': 3, 'reading': 2, 'teacher': 2, 'writing': 1, 'learning': 1,
        'teaching': 1})

Enter sentence: the student is reading

Top 5 Next Words:
the = 0.25
is = 0.25
student = 0.15
reading = 0.1
teacher = 0.1

Unseen N-gram example:
PS C:\Users\reddy\nlp\DSA0304-NATURAL-LANGUAGE-PROCESSING>
```

2.

Enter sentence: the student likes python

UNSMOOTHED

the -> 0.0

student -> 0.0

is -> 0.0

reading -> 0.0

a -> 0.0

BACKOFF

the -> 0.2121

student -> 0.1515

is -> 0.1515

a -> 0.0909

reading -> 0.0606

INTERPOLATION

the -> 0.0424

student -> 0.0303

is -> 0.0303

a -> 0.0182

reading -> 0.0121

PS C:\Users\reddy\nlp\DSA0304-NATURAL-LANGUAGE-PROCESSING>

3.

```
er" 60653" -- "C:\Users\reddy\nlp\DSA0304-NATURAL-LANGUAGE-PROCESSING\CO3_AT1\Q4.py"
Unigram Entropy: 3.0564
Bigram Entropy: 0.9406
Trigram Entropy: 0.6462
PS C:\Users\reddy\nlp\DSA0304-NATURAL-LANGUAGE-PROCESSING>
```

4.

```
er" 60717" -- "C:\Users\reddy\nlp\DSA0304-NATURAL-LANGUAGE-PROCESSING\CO3_AT1\Q4.py"
Enter an English sentence: a student writing python

Rule-Based Tagger:
[('a', 'DT'), ('student', 'NN'), ('writing', 'VBG'), ('python', 'NNP')]

Stochastic Tagger:
[('a', 'DT'), ('student', 'NN'), ('writing', 'VBG'), ('python', 'NNP')]

Transformation-Based Tagger:
[('a', 'DT'), ('student', 'NN'), ('writing', 'VBG'), ('python', 'NNP')]
PS C:\Users\reddy\nlp\DSA0304-NATURAL-LANGUAGE-PROCESSING>
```